~/paulynomial

paul kiernan

staff+ infrastructure engineer — search, retrieval & streaming systems

hi@paulynomial.com 646.369.6726 paulynomial.com github.com/paulkiernan linkedin.com/in/paulynomial San Francisco, CA

summary

Fifteen years building search and streaming infrastructure at scale. At Reddit (500M+ monthly users), led search indexing for a multi-billion-document Solr/Lucene index, shipped comment search (7.5M+ new comments a day, searchable within 60 seconds), and designed the 5M+ messages/sec Kafka platform other teams built on; the same systems later served Reddit's LLM products. Founding engineer, then CTO, at Phaselaw: built an AI redaction platform from zero to SOC 2 Type II and 10M+ documents for the NHS and others.

experience

Sabbatical
Planned break after Phaselaw — San Francisco, CA — Nov 2025 – Present

- Time with family, and digging into how retrieval infrastructure changes when AI agents, not people, are the ones searching.

Phaselaw
AI document review and redaction — Remote — Jul 2023 – Nov 2025

Chief Technology Officer — Oct 2024 – Nov 2025

- **Platform.** Ran engineering for AI redaction of highly sensitive legal and personal data, trusted with 10M+ documents by the NHS, the UK Labour Party, and the Raspberry Pi Foundation.
- **Team.** Built and led the engineering organization: hiring, culture, and technical due diligence with investors and partners.
- **SOC 2 Type II.** Took the platform through certification: CI/CD, automated testing, OpenTelemetry, and multi-region Terraform on AWS keeping customer data in the US, EU, or UK.
- **AI pipeline.** Claude on AWS Bedrock finding PII in legal documents, orchestrated by Temporal on multi-tenant Kubernetes.

Founding Backend Engineer — Jul 2023 – Oct 2024

- **Built from zero.** Designed and shipped the core backend: data models, API, auth, and a service architecture spanning multiple global regions.

Reddit
500M+ monthly users — San Francisco, CA — Jul 2018 – Jul 2023

Staff Software Engineer, Tech Lead, Search Indexing — Nov 2019 – Jul 2023

- **Search indexing.** Led the team keeping a multi-billion-document Solr/Lucene index fresh against millions of events per second; managed 3 engineers, mentored 3 more, and designed the Scala/Flink pipeline.
- **Comment search.** Gave Reddit comment search after a decade without it: a Kafka consumer DAG that makes 7.5M+ new comments a day searchable within 60 seconds of posting.
- **Kafka platform.** Designed the Kafka platform that finally scaled for Reddit: a 5M+ messages/sec backbone, adopted by analytics, data warehousing, and notifications.
- **LLM retrieval.** These indexing and retrieval systems went on to serve all of Reddit's LLM applications.
- **Reliability.** Led search incident response and materially improved Reddit Search uptime.

Senior Software Engineer II, Data Infrastructure — Jul 2018 – Nov 2019

- **Real-time warehouse.** Cut warehouse latency from hours to seconds, replacing cron batch jobs with event-driven Kafka-to-BigQuery pipelines.

Chartbeat
Real-time analytics for publishers — New York, NY — Aug 2014 – Jul 2018

Head of Backend Engineering — Oct 2016 – Jul 2018

- **Leadership.** Ran backend engineering: 8 engineers across 4 product teams. Before that, led the R&D team and presented its experimental products to the board.
- **Systems.** Built MrRoboto, a chain-replicated C++ key-value store taking 500k+ writes/sec; Miata, a real-time streaming analytics framework; and the probabilistic counting behind the "so far today" API. Open-sourced Parselmouth.

Moat
Ad analytics, acquired by Oracle — New York, NY — Jun 2012 – Aug 2014

Software Developer

- **Ad intelligence.** Early engineer on Moat Pro, tracking online ads for enterprise clients: built the geo-distributed crawler network, ETL pipeline, and reporting API that captured and indexed them.

skills

languages	Python, Scala, C++, Java, C, Clojure, SQL, Shell
infrastructure	Kafka, Apache Flink, Solr/Lucene, BigQuery, Temporal, ZeroMQ, AWS (multi-region), Kubernetes, Terraform, ArgoCD, OpenTelemetry, HAProxy
ai / llm	LLM extraction pipelines (Claude on AWS Bedrock), retrieval infrastructure for LLM applications

education & patents

education	**Cornell University**, BS in Electrical & Computer Engineering, 2012. McMullen Dean Scholar; Goldfarb Tradition Fellow.
patent	**US 10,068,250**: "System and method for measuring mobile advertising and content by simulating mobile-device usage."